



# JESÚS DANIEL CABALLERO COLINA

## MACHINE LEARNING ENGINEER

### ML Systems | AI Applications | Edge Deployment

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#### CONTACT:

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#### PROFESSIONAL SUMMARY:

Machine Learning Engineer focused on building production-grade AI and ML systems, with experience developing end-to-end applications integrating backend services, data pipelines, ML models, and edge deployment. Specialized in Python, FastAPI, and Docker for developing forecasting, anomaly detection, and time-series analysis solutions.

Currently leading the design and development of AI applications within the Proctek IA ecosystem, including predictive monitoring systems, backend APIs, AI agents, and inference platforms deployed on ARM-based edge architectures using Docker/Podman. Interested in roles related to AI Engineering, Machine Learning Engineering, and backend architecture for ML systems.

#### PROFESSIONAL EXPERIENCE:

##### Proctek S.A. – Data Analytics Engineer

- Led the design and development of AI and machine learning applications focused on time-series analysis, anomaly detection, and forecasting over operational data.
  - Built end-to-end systems integrating FastAPI backend services, data pipelines, machine learning models, and interactive dashboards for real-time monitoring and analytics.
  - Designed inference and deployment architectures for containerized AI applications using Docker/Podman on ARM-based edge devices, optimized for continuous operation and limited-connectivity environments.
- March 2024  
Present
- Implemented machine learning functionalities including health scoring, anomaly detection, multivariate forecasting, and time-series data processing for production-deployed applications.
  - Contributed to the development of AI agents for technical and operational dataset analysis, integrating tool orchestration workflows and local context retrieval capabilities.
  - Defined technical architecture decisions, development standards, and deployment strategies, while providing technical guidance to junior engineers during feature development.
  - Led technical validations and solution demonstrations with clients, translating operational requirements into functional and deployable AI applications.

##### Quynza Technologies – Mechatronics Engineer

- Developed mechatronic solutions integrating hardware and software for automation and data acquisition systems.
  - Programmed embedded systems based on Arduino, ESP32, and MicroPython for IoT and control applications.
  - Contributed to the development and digitalization of technical training courses focused on IoT, MicroPython programming, and digital logic, integrating interactive content into the company's web platform.
- March 2022  
January  
2024

#### EDUCATION:

##### **MSc in Industrial Automation Engineering, 2023 – 2025**

National University of Colombia, Bogotá D.C.

*Thesis focused on predictive analytics and industrial asset modeling using AI/ML software for Oil & Gas applications.*

##### **BSc in Mechatronics Engineering, 2017 – 2022**

National University of Colombia, Bogotá D.C.

## SKILLS:

- Programming Languages: Python, SQL, C++, Java
- Machine Learning & AI: scikit-learn, time-series forecasting, anomaly detection, feature engineering, AI agents, RAG
- Backend & APIs: FastAPI, REST APIs, backend architecture for AI systems
- Infrastructure & Deployment: Docker, Podman, GitHub, edge deployment on x86/ARM architectures
- Data: pandas, NumPy, SQLite, time-series processing
- Industrial & Edge Systems: IIoT, ThingWorx, ESP32, PLCNext

## PROJECTS:

### Project 1 - Predictive Analytics and Energy Forecasting

- Developed predictive models and time-series analysis workflows for energy market and electricity pricing datasets, applying statistical and machine learning techniques for forecasting and temporal pattern analysis.
- Built preprocessing, feature engineering, and modeling pipelines in Python using SARIMAX, Prophet, AutoARIMA, PCA, DBSCAN, and K-Means over hourly time-series and multivariate datasets.

### Project 2 - AI-Based Predictive Maintenance System

- Developed anomaly detection, condition classification, and forecasting models for pumping systems using high-frequency multivariate industrial datasets.
- Explored and evaluated deep learning architectures for industrial time-series analysis, including LSTM, Transformers, and Autoencoders, together with preprocessing and feature engineering pipelines for industrial sensor data.

### Project 3 - ML Analysis and AI Pipeline Execution Platform

- Developed an ML platform for structured analysis of tabular and time-series datasets, focused on dataset readiness validation, feature engineering, and reproducible baseline modeling.
- Designed a backend architecture using Python and FastAPI for reproducible ML pipeline execution, integrating artifact-based processing and optional LLM/RAG capabilities.

## CERTIFICATIONS:

|      |                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>Professional Certificate - IBM Machine Learning</b><br>Coursera – IBM                                                           |
| 2025 | <b>Professional Certificate - IBM Data Science</b><br>Coursera – IBM                                                               |
| 2025 | <b>Specialization “IoT Systems and Industrial Applications with Design Thinking”</b><br>Coursera – LearnQuest                      |
| 2024 | <b>Online Course “ThingWorx Technical Essentials (Self-Paced)”</b><br>PTC Field Academy                                            |
| 2023 | <b>Specialization “An Introduction to Programming the Internet of Things (IOT)”</b><br>Coursera – University of California, Irvine |
| 2020 | <b>Specialization “Mathematics for Machine Learning”</b><br>Coursera – Imperial College London                                     |

## LANGUAGES:

|          |                                           |
|----------|-------------------------------------------|
| SPANISH: | Native                                    |
| ENGLISH: | B2 (Professional technical communication) |

## HONORS:

- **National Recognition – Saber Pro Exam (2022):** Awarded by the Colombian Ministry of Education for achieving top performance in the national standardized exam for engineering graduates.
- **Best Undergraduate Thesis – Mechatronics Engineering (2022):** Awarded by the National University of Colombia for outstanding academic and technical contribution in undergraduate research.
- **Top Academic Performance – Mechatronics Engineering Cohort (2023):** Recognized by the National University of Colombia among the highest-performing graduates of the 2022-2 cohort.